

Original Manuscript ID: Access-2025-32935

Original Article Title: *“Performance Comparison of Beamsteering Approaches for Coverage Enhancement in Indoor Optical Wireless Communications”*

To: IEEE Access

Dear Prof. Yogendra Prajapati,

Thank you for your decision on our manuscript (email received on August 25, 2025) and for your recommendations for its re-submission.

We have carefully revised our paper according to the comments received from the reviewers, whom we would like to thank for their constructive work.

You will find more details on the modified parts in the “Response to the Comments”. Also, all modified parts in the paper are blue colored.

We believe the paper is now much improved thanks to these comments and suggestions.

Please notice that in the response parts, all reference, figure, equation, and section/subsection numbers refer to the revised paper.

In addition to the detailed responses provided to the reviewers in the following section, the authors would like to highlight here the main changes made to the new version:

- **The addition of a complete section (Section IV) devoted to experimental results; the first version only included modeling and simulation work.**
- **Simplification of the original simulation section to make it clearer and more concise (removal of results in cases where the source power is fixed at $P = 1$ W and merging of all simulation results in a dedicated Section III).**

We therefore believe that these major changes strengthen the article.

Thank you very much for your attention.

Looking forward to hearing from you.

Bastien Béchadergue and co-authors

**Responses to the Reviewers' Comments on the manuscript with ID
Access-2025-32935**

***Performance Comparison of Beamsteering Approaches for Coverage
Enhancement in Indoor Optical Wireless Communications***

by Haitham AL Satai, Bastien Béchadergue and Luc Chassagne

February 4, 2026

We would like to thank the reviewers and the editor for their constructive comments. They have been very helpful in the revision of this paper and allowed us to improve the technical content and presentation quality. We thus hope that we have taken into consideration and complied with all the comments and suggestions.

In this document, we provide detailed explanations of the changes made in the revised version of our article to address these comments. Each comment is reminded in **red** and is followed by our response, in black font. The changes made to the revised manuscript are mentioned in **blue**. Please also note that, unless stated otherwise, all page, section, figure, table, and reference numbers, as well as acronyms refer to the revised manuscript.

Answers to Reviewer 1

The manuscript is focused on the impact of beamforming/beamsteering in OWC. In this sense, the topic is interesting for IEEE ACCESS. However, the main issue is that the manuscript considers several topics; beam management, reflectors, modulations, etc. but it does not dive deep in any of these topics. That is, it can be considered a mere extension of [25] just adding some specific topics that does not contribute to the novelty of the paper.

We thank the reviewer for these comments, which have enabled us to significantly improve the article. Thanks to the major changes listed in the cover letter, and discussed in more detail later in this document, namely the addition of a complete experimental section (Section IV) and more concise developments in the simulation section (Section III), we believe that this article is not a simple extension of [25] but builds upon this previous work to provide new results and insights.

To clarify the contributions of this article, we have updated its abstract and introduction (Section I) with an explicit comparison with [25]. We have also included the new experimental results in the new Section IV.

- 1) The presented manuscript relies on the concept of beamforming. However, it is not clear how these beams are generated. Are the beams generated through lenses or VCSELs are considered? In the latter case, notice that the Lambertian propagation may not completely fit to the signal propagation.

It is absolutely true that the type of source used was not explicitly mentioned, particularly during the system and channel modeling stage (Section II). In this work, we do not use VCSELs but infrared (IR) light-emitting diodes (LEDs), whose emission pattern is assumed to be Lambertian. In the simulation sections (Sections II and III), we consider that this LED is not coupled to any other optical component and that the emission pattern is indeed Lambertian (i.e., governed by the

equations (3) and (4)). In the new section of the article devoted to experimental results (Section IV), we verified this hypothesis with the LED mounted in the commercial LiFiMAX OWC system we used. As shown in Figure R1 below, this LED (consisting of an LED chipset coupled to a silicone dome) follows the red measured pattern, which is quasi-Lambertian, as can be concluded after comparison with the theoretical Lambertian pattern in black (the green curve corresponds to a polynomial fitting of the measured pattern not used here).

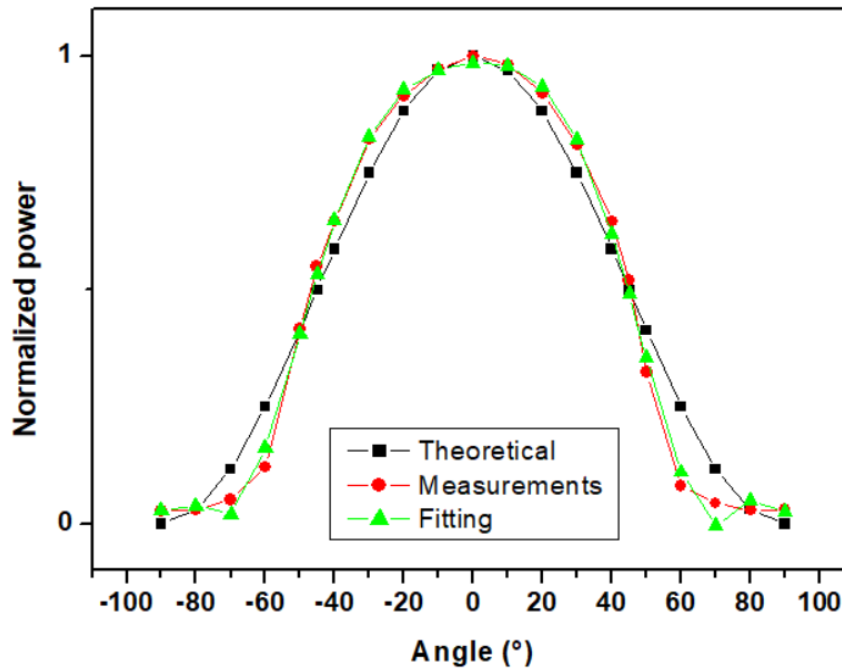


Figure R1 – Beam pattern of the LED used in the experimental section (Section IV) of the revised article.

For the sake of brevity, and because this verification is already included in another article from the same authors (see reference [37] added to the revised manuscript), we have not included this figure in this new version of the article. However, we have added a paragraph in Section II-B, just after equation (1), to clearly state that we focus of IR LED and justify this choice (see the answer to Comment 2 from Reviewer 1 below for more details about this justification) and have mentioned in Section IV-A that the IR LED used in the experimental work has a close-to-Lambertian pattern, citing [37] as a justification.

2) The transmission wavelength corresponds to infrared domain. Justify why and what LED or VCSEL technology is considered.

As indicated in response to the previous comment, we do not use VCSELs in this work, but rather LEDs, which do indeed emit infrared light. There are several reasons for this choice:

- First, the test bench presented in the new experimental section (Section IV) was developed with an infrared LED rather than a VCSEL or laser source for photobiological safety reasons and because LEDs offer wider coverage due to their lower directivity.
- Furthermore, we chose an infrared wavelength rather than a visible one because the optical source is coupled to a beamsteering system. A steerable visible source would generate a moving light flux visible to users, which is undesirable for obvious reasons of ambient lighting uniformity.

In order to clarify this hypothesis of an IR LED source and justify it, we have added a paragraph rephrasing the above explanations in Section II-B, just after equation (1).

In parallel, it should be noted that simulations similar to those presented in this article were also carried out with a laser source (modeled as having a Gaussian emission pattern), but we believe that this article is already quite long and that we cannot include them, given that under the same signal assumptions (signal bandwidth and noise level in particular), the trends observed with a Gaussian laser are similar to those reported in the article with a Lambertian LED. At the same time, the case of a laser or VCSEL source transmitting a signal of larger bandwidth (e.g. in the range of a few hundreds of MHz) would require changing the channel model to account for delay spreads, which is out of scope of this article (see also answer to Comment 3.c from Reviewer 2).

These justifications are provided at the end of Section II-B-1) in order to explain why we are only considering IR LEDs, while opening the discussion to these other types of optical sources.

- 3) The NLoS model corresponds to specular or diffuse reflections? In Fig. 1a seems that the authors are considering mirror or some reflecting material that may improve the received signal.

In Figure 1(a), the squares called ‘reflectors’ are not mirrors (i.e., they do not produce specular reflections) but elementary wall surfaces, which are therefore modeled as materials that induce diffuse reflections of the light to which they are exposed. In accordance with the literature on the subject, these elementary surfaces are modeled as passive Lambertian sources of order 1 with a specific reflectivity, generally 0.6, as explained in [32].

We then couple this modeling with the frequency-based approach proposed by Schulze in [27] in order to simulate indirect paths more quickly but nevertheless exhaustively compared with the time-based approaches commonly used in the literature (the reflections taken into account are thus not limited to order 1 or 2 but extend to infinite order).

However, it should be noted that the accuracy of indirect path modeling and simulation time depend heavily on the area dA of these elementary surfaces, so the value of this parameter must be chosen carefully, as specified in Section III-A.

To avoid any confusion about reflectors and to clarify the point raised by the reviewer, we have modified the text in Section II-B just above equation (5).

- 4) A scenario covering an area of 20m x 20 m is considered deploying a single AP. This assumption involves the lack of interference; however, this area typically requires multiple AP. Justify the deployment of a single AP.

The aim of this article is precisely to show that an OWC system with a single steerable optical source at the access point (AP) level is capable of covering a larger area than conventional systems based on fixed optical sources, each generating a fixed cell. The simulations and experiments carried out are intended to quantify the coverage gain achieved by beamsteering. This is why we chose a large area of 20 m x 20 m. These very large dimensions ensure that in most simulated cases (but not all, as suggested by the simulation results with OOK modulation), the coverage obtained will be less than 400 m², allowing us to observe the communication coverage

provided by a single steerable AP. We have reworded the beginning of Section III-A to justify this assumption.

5) The noise $n(t)$ is not defined. Is it thermal noise, shot noise, ambient noise?

In order to model noise in the simulations, we followed standard practices in the OWC-related literature, i.e., we considered that the noise at the OWC receiver consists of both thermal noise and shot noise.

In terms of shot noise, we neglected the contribution of the received data signal, which, according to the literature, is generally much lower than the shot noise caused by ambient light (please also refer to the answer to Comment 3.a from Reviewer 2 for further details about this assumption).

Under these conditions, thermal noise and shot noise can be modeled as additive white Gaussian noise (AWGN) sources, the sum of which is also an AWGN source. We therefore included all these contributions in the variable N_0 and adopted a value for this variable taken from literature.

To clarify this point, we have added a dedicated paragraph in Section III-A.

6) The impact of misalignment can be minimized applying the concept of angle diversity receiver (ADR).

We fully agree with the reviewer. However, deploying such an approach through simulation and experimentation is not the subject of this article, which focuses instead on quantifying the gain in communication coverage obtained by using an optical beamsteering system at a single OWC AP. Nevertheless, we have mentioned this solution through references on the subject when commenting on the simulation results of the case with misalignment (see end of Section III-C and references [35] and [36]) to bring this topic to the attention of readers.

In general, the paper has potential to be published in IEEE ACCESS. However, it mixes several concepts that should be properly clarified and organized to generate an interesting contribution. I encourage to resubmit the manuscript once all these issues will be solved.

We would like to thank the reviewer for all of these constructive comments and hope that incorporating them into this new version of the article has improved its quality.

Answers to Reviewer 2

In this paper the authors simulate an indoor OWC downlink in a $20 \times 20 \times 2.5 \text{ m}^3$ room with a ceiling-mounted IR LED access point (AP) at 940 nm and a user equipment (UE) moving on a plane 0.85 m above the floor (1.65 m below the AP). They compare five scenarios: (i) with beamsteering (ideal continuous alignment), (ii) without beamsteering (fixed vertical AP/UE), (iii) manual orientation (UE aligns to a fixed AP), (iv) source with various fixed orientations (multi-source fixed tilts), and (v) misalignment (small steering errors). Coverage is defined as the area where $\text{BER} < 3.8 \times 10^{-3}$. Three modulations are considered: OOK, DCO-OFDM, and m-CAP. NLOS reflections are modeled with the frequency-domain approach of Schulze (matrix formulation).

Claimed additions vs. authors' earlier work: The authors extend their prior LOS-only study ([25]) by: (1) adding two scenarios—fixed multi-orientation sources and misalignment; (2) including NLOS reflections; and (3) comparing OOK and DCO-OFDM to m-CAP. This is summarized in the Introduction and Sec. II-A and revisited across Secs. III–IV.

We would like to thank the reviewer for the following stimulating comments, which we hope we have addressed comprehensively in order to improve the quality of our article.

The reviewer has the following concerns:

- 1) Novelty and scope: The paper does not propose a new beamsteering method, control policy, algorithm, or analytical coverage formula. It is a parameter study with Monte-Carlo-like grid evaluations of a fairly standard channel model. The new scenarios (misalignment; fixed multi-source tilts) are incremental and the main qualitative conclusions (beamsteering helps; OOK more robust but low spectral efficiency; reflections can help) are expected. For a high-rank journal, this needs either analytical advances, a strong optimization/design contribution (e.g., provably optimal orientation sets for a given privacy/coverage target), or measurement-validated results that reveal non-obvious trade-offs.
 - a. The core premise—that beamsteering improves coverage in OWC—is not novel. It is well-established in the literature (as partially referenced by the authors, e.g., [17]–[24]) and is intuitively obvious. The paper confirms this known fact with more parameters but does not challenge or fundamentally advance the underlying theory.
 - b. The extension from the prior conference paper [25] (which only used m-CAP) to include OOK and DCO-OFDM is a logical and incremental step, not a transformative one. The results (OOK has better coverage but lower data rate, m-CAP is better than OFDM due to PAPR) are entirely expected from communication theory and do not constitute a novel discovery.
 - c. The two new scenarios ("misalignment" and "fixed orientations") are the strongest candidates for novelty. However, the analysis of misalignment, while useful, remains relatively superficial. The "fixed orientations" scenario is essentially a study of a multi-beam static access point, a concept that is also known in the field (e.g., as a way to create atto-cells without mechanical steering).

We thank the reviewer for this insightful comment, which led to a considerable rewriting of the article. The authors agree that beamsteering is well known to improve peer-to-peer communications. However, we believe that literature lacks a quantitative approach that clearly evaluates the communication performance gains provided by an OWC system with beamsteering compared with a system without beamsteering.

To this end, two approaches could have been used: a purely analytical approach consisting of mathematically deriving the link between communication metrics (e.g., BER, SNR, etc.) and transmitted data signal parameters; and a Monte Carlo approach based on end-to-end simulations of the communication system in order to evaluate these same communication metrics following the repeated generation of random experiments (randomness occurring in particular at the level of the transmitted binary frame and reception noise).

We ultimately opted for the second approach, as we wanted to include in our analysis the indirect paths resulting from reflections of the optical signal on the environment, which the analytical approach would not have allowed us to do satisfactorily. It should be noted that the consideration of these non-line-of-sight (NLOS) paths constitutes a major development compared with [25], in addition to the consideration of new modulations and new scenarios as mentioned by the reviewer. The proposed simulation model therefore allows us to take into account a very large number of input parameters in a more flexible and usable way than an analytical approach would allow. In this respect, we believe that the comprehensive Monte Carlo approach developed and adopted is in itself a contribution that should be of interest to readers from the OWC community, and that is no less legitimate than an analytical approach, even for a high-ranking journal.

However, to supplement this initial contribution, we have included in this revised version of the article a new section (Section IV) devoted entirely to the experimental study of the coverage gain achieved by an OWC system with beamsteering. We detail the setup implemented and used, and compare the data rate, SNR, and coverage performance obtained in scenarios ‘with beamsteering’, ‘with manual orientation’, and ‘without beamsteering’, in order to eventually provide high-level comparison between the experiments and the simulations.

In order to clearly highlight the contributions of this article, particularly in relation to [25], we have reworked the introduction (Section I) as indicated in response to Comment 1 from Reviewer 1. In addition, we have added our new experimental results in the new Section IV.

- 2) Photobiological safety modeling (Eq. 6) is physically inconsistent: In Sec. II-C (p. 7), the paper states that the average transmitted optical power must satisfy a “photobiological safety” limit $P_{t,max}(\Phi_{1/2})$ and then gives Eq. (6):

$$P_{t,max}(\Phi_{1/2}) = \frac{8\pi \cos \Phi_{1/2}}{\ln(\cos(\Phi_{1/2})) - \ln(2)}$$

This expression yields negative power for practical angles (because $\ln(\cos(\Phi_{1/2})) < 0$ and $\ln(2) > 0$), e.g., $\Phi_{1/2} = 60^\circ \Rightarrow P_{t,max} \approx -9.06 \text{ W}$; $\Phi_{1/2} = 45^\circ \Rightarrow \approx -17.1 \text{ W}$ —nonsense for a power limit. Yet the text later claims “ $P_t = 1 \text{ W}$ is excluded for small angles” and that $P_{t,max}$ “reaches $\sim 12.5 \text{ W}$ at 60° ” (Sec. III-C, p. 10), which contradicts Eq. (6). This is a critical error: the safety model is neither derived nor consistent with IEC 62471’s radiance/irradiance exposure rules, which depend on wavelength, exposure time, source angular subtense, pupil size, etc.—not solely on $\Phi_{1/2}$ by a simple log-cosine formula. The

entire set of “ $P_t = P_{t,max}$ ” results is therefore on unreliable footing and must be redone with a correct, documented safety model.

We thank the reviewer for pointing out this error, for which we apologize and which we have of course corrected in the revised version of the article. This error was in fact a typographical error made when writing (8) (formerly (6)) in the manuscript (missing logarithm term in the numerator). The formula used in the simulations was however correct, so the maximum powers allowed in these simulations remain unchanged, as do the results of these simulations.

Furthermore, it is true that the IEC-62471 standard specifies different methods depending on wavelength, exposure time, source angular subtense, pupil size, etc. in order to determine whether an LED source is safe from a photobiological point of view. However, in the case of a near-infrared point source (more precisely between 780 nm and 1400 nm), which is the case considered in this article, an LED source is safe if its irradiance along the optical axis at 20 cm from the source is below an irradiance limit E_{max} , given by:

$$E_{max} = \begin{cases} 18000 \times t_{exp}^{-0.75} \text{ W} \cdot \text{m}^{-2} & \text{when } t_{exp} \leq 1000 \text{ s,} \\ 100 \text{ W} \cdot \text{m}^{-2} & \text{when } t_{exp} > 1000 \text{ s.} \end{cases} \quad (\text{R1})$$

By adhering to these guidelines, the IR source remains safe for both short and long-term exposure durations. In this article, we have considered the worst-case scenario in order to determine the maximum transmit optical power allowed $P_{t,max}(\Phi_{1/2})$ as a function of the semi-angle at half power $\Phi_{1/2}$, i.e., the case when $t_{exp} > 1000 \text{ s}$, so that $E_{max} = 100 \text{ W/m}^2$.

However, according to the propagation model for Lambertian LED, we can deduce from the equation (3) in the article that the irradiance $E(d, \theta, \phi)$ at any point of spherical coordinates (d, θ, ϕ) is given by:

$$E(d, \theta, \phi) = E(d, \phi) = \frac{P_t(m+1)}{2\pi d^2} \cos \phi. \quad (\text{R2})$$

Knowing that $m = -\ln 2 / \ln(\cos(\Phi_{1/2}))$ and setting the equality $E(0.2, 0) = 100 \text{ W/m}^2$ (i.e., the LED irradiance along the optical axis at 20 cm from the source equals 100 W.m^2), we can deduce that the maximum transmit optical power allowed $P_{t,max}(\Phi_{1/2})$ is:

$$P_{t,max}(\Phi_{1/2}) = \frac{8\pi \ln(\cos(\Phi_{1/2}))}{\ln(\cos(\Phi_{1/2})) - \ln(2)}. \quad (\text{R3})$$

To clarify how this equation is obtained, we have significantly revised former Section II-C, dividing it into two subsections, the first detailing the waveform generation process (Section II-C.1, see answer to Comment 3.b from Reviewer 2 below for more details) and the second (Section II-C.2) detailing the procedure followed to determine the maximum acceptable average optical power presented above. This new subsection also discusses the implications of the coupling of this power limit with the waveform generation process, as further detailed in the answer to Comments 3.b and 7.e from Reviewer 2.

3) Noise and channel modeling is oversimplified and biases conclusions:

- a. The receiver noise is modeled as constant-PSD AWGN with $N_0 = 10^{-21} \text{ W/Hz}$ (Table III, p. 8) and no signal-dependent shot noise nor ambient illumination noise

at 940 nm. Real indoor OWC links are usually shot-noise limited when ambient light is present; ignoring this strongly overestimates coverage and advantages OOK in particular.

We thank the reviewer for this comment on noise modeling, which echoes Comment 5 from Reviewer 1. As mentioned in the answer to this comment, we modelled the noise following standard practices in the OWC literature, i.e., we considered that the noise at the OWC receiver consists of both thermal noise and shot noise.

Regarding shot noise, we have neglected the contribution induced by the received data signal, as pointed out by the reviewer. We have adopted this assumption, which is very common in OWC literature [11], because the propagation distances in our simulations are such that the received optical power corresponding to the data signal is very low compared with that corresponding to the ambient lighting. Consequently, the signal-dependent shot noise is negligible compared with that from ambient lighting.

However, contrary to what the reviewer suggests, we have not neglected this ambient lighting-induced shot noise, which, according to OWC literature, can be modeled as AWGN [11]. At the same time, thermal noise can also be modeled as AWGN, so that the sum of these two noise sources is also AWGN. We therefore included all of these contributions in the unique variable N_0 and adopted a value for this variable taken from the literature (e.g., [34]).

To clarify this point, we have added a dedicated paragraph in Section III-A.

- b. The paper later attributes m-CAP's advantage over OFDM to PAPR (Sec. III-B, p. 9), but no nonlinearity/clipping or peak-current limit is actually modeled; with a pure AWGN model at a fixed average optical power, PAPR does not confer a BER advantage. The current explanations are inconsistent with the simulation assumptions.

In order to clarify why we use PAPR to explain the difference in coverage between *m*-CAP and OFDM modulation, it is necessary to clarify how the transmitted signal is constructed. Whether using *m*-CAP, OFDM and also OOK, the optical data signal is simulated according to the following steps:

- 1st step: A random binary stream of N_b bits is generated (with $N_b = 10^5$).
- 2nd step: This binary stream is modulated, either using OOK, *m*-CAP or OFDM. Let us denote the corresponding discrete waveform $\{x_n\}_{1 \leq n \leq N_s}$, where x_n is the n -th data sample and N_s the total number of samples in the waveform.
- 3rd step: In parallel, we modeled the LED as an electro-optical converter that transforms these input waveform samples into corresponding optical power levels $\{P_{t,n}\}_{1 \leq n \leq N_s}$ according to a constant conversion factor. Such an assumption is a good first order approximation, given that the frequency response of the LEDs considered (including in the experiments) is constant over the frequency range of the transmitted signal (here 30 MHz, see answer to Comment 3.c from Reviewer 2 for more details). We also defined the maximum allowed average optical power, denoted $P_{t,\max}$, depending on the considered

LED parameters, following the procedure previously described in the answer to Comment 2 of Reviewer 2.

- 4th step: In the case of OOK, we then applied Manchester coding, which ensures a balance between high and low states, and then assigned the samples corresponding to the high states to the optical power level $2P_{t,\max}$, and the low states samples to a null optical power (i.e., LED off). This eventually guarantees an average power equal to $P_{t,\max}$.
- 5th step: In the cases of DCO-OFDM and m -CAP, we also ensured the transmitted optical power had an average value of $P_{t,\max}$. However, if the waveform $\{x_n\}_{1 \leq n \leq N_s}$ is in both cases not rescaled or clipped first, some samples might fall out of the linear range of the LED response. In this work, we did not use clipping but instead rescaled the samples so that they eventually fell in the range $[-P_{t,\max}/2, P_{t,\max}/2]$ after optical conversion. This ensured that after a DC bias $P_{t,\max}$ is added to the rescaled waveforms, all the samples of the LED drive signal eventually fell in the interval $[\frac{P_{t,\max}}{2}, \frac{3P_{t,\max}}{2}]$ after optical conversion, i.e., remained positive and in the linearity range of the LED, set between $P_{t,\max}/2$ and $3P_{t,\max}/2$ (see below for a discussion about this range). In practice, this was done using the relationship:

$$P_{t,n} = \frac{x_n}{\max\{|x_n|\}_{1 \leq n \leq N_s}} \times \frac{P_{t,\max}}{2} + P_{t,\max}. \quad (\text{R4})$$

This waveform generation process has three main implications:

- Although the waveforms have the same average optical power regardless of the modulation considered, OOK allows the transmission of a higher instantaneous optical power ($2P_{t,\max}$ against $3P_{t,\max}/2$), which is a deliberate choice that makes the best use of the LED's available optical power.
- Scaling the DCO-OFDM and m -CAP waveforms within the linear range of the LED avoids nonlinearity and therefore clipping. This may not be optimal but allows to exclude this factor when comparing the performance of both modulations.
- Finally, and more importantly with respect to the comment of the reviewer, both m -CAP and OFDM waveforms are known to follow Gaussian distributions of zero mean and specific variances, with m -CAP having a lower peak-to-average power ratio (PAPR) than DCO-OFDM [29]. This means that with the waveform rescaling process described above, the variance of the transmitted optical signal $\{P_{t,n}\}_{1 \leq n \leq N_s}$ is larger with the former, and so is the variance of the corresponding received signal. The received data signal is indeed overall equal to $R_{PD}P_{t,n}(H_{LOS} + H_{NLOS})$, with H_{LOS} and H_{NLOS} the LOS and NLOS channel gains given by (3) and (5) and R_{PD} the photosensitivity of the photodiode, so the statistical distribution of the received signal is overall that of the transmitted signal. Given that the variance of the received signal corresponds to the useful received power for proper data reception, we can deduce that the signal-to-noise ratio (SNR) with m -CAP should be higher than that obtained with DCO-OFDM in the same setup (i.e., same geometrical configuration and same noise statistical distribution). This eventually explains why the coverage performance with m -CAP is overall better than with DCO-OFDM.

In order to clearly justify this difference in PAPR, we have significantly revised Section II-C, dividing it into two subsections, the first repeating the explanation of the waveform generation process above (Section II-C.1) and the second (Section II-C.2) detailing the procedure followed to determine the maximum acceptable average optical power described in the answer to Comment 2 from Reviewer 2 above.

It may be noted that this waveform generation process coupled with the maximum allowed average optical power constraints obtained with (8) has two additional implications:

- One could argue here that there is no guarantee that the response of a single LED will be linear over the range $\left[\frac{P_{t,\max}}{2}, \frac{3P_{t,\max}}{2}\right]$, particularly for diffuse LEDs for which the maximum power allowed by the IEC-62471 standard is very high (e.g., 12.5 W if $\Phi_{1/2} = 60^\circ$). In practice, however, we consider that to implement such a transmitter, it is anyway not possible to use a single LED chipset, but that several LEDs driven each in their linearity range must be used in parallel to transmit the same signal simultaneously and thus add up their optical power.
- It should finally be noted that, according to (R4) (i.e., (6) in the revised article), the optical power emitted may occasionally exceed the photobiological limit $P_{t,\max}$. However, the average transmitted optical power is eventually equal to $P_{t,\max}$, as the data signal $\{x_n\}_{1 \leq n \leq N}$ has zero mean. In addition, the duration during which the optical power exceeds $P_{t,\max}$ is very short, lasting only a few sampling periods at most. Over such a short period, the photobiological threshold no longer corresponds to a maximum irradiance of 100 W/m², but is much higher, as indicated by equation (R1) (i.e., (7) in the revised article), meaning that these exceedances do not ultimately pose any photobiological safety issues.

In order to clarify these implications for the reader, we have added the above elements to the new Section II-C.2 dedicated to determining the maximum allowed average optical powers.

- c. Channel: The text asserts the channel is flat up to “a few tens of MHz” (Sec. II-B, p. 6–7) and collapses the impulse response to a scalar gain in Eq. (2), yet the simulations use bandwidths up to 30 MHz (DCO-OFDM) / 15 MHz (m-CAP sub-bands), where multipath ISI often matters in rooms of this size. A true frequency-selective response (which the authors already have via the Schulze method) should be used to filter the waveforms, otherwise OOK’s “coverage” gains are inflated and the relative behavior of OFDM vs. m-CAP is not trustworthy.

We thank the reviewer for this comment. Assumptions regarding the propagation channel do indeed have direct consequences on communication and coverage performance, and must therefore be duly justified.

In this work, we consider that regardless of the modulation used, the bandwidth exploited ranges from 1 MHz to 30 MHz. This assumption is mainly motivated by the measurements of the exploitable bandwidth of the optical antennas we used in the experimental part (Section IV). Using a vector network analyzer, we obtained the normalized frequency response shown in Figure R2 for the system consisting of a transmitting optical antenna, indoor propagation (including reflections off walls), and a receiving optical antenna.

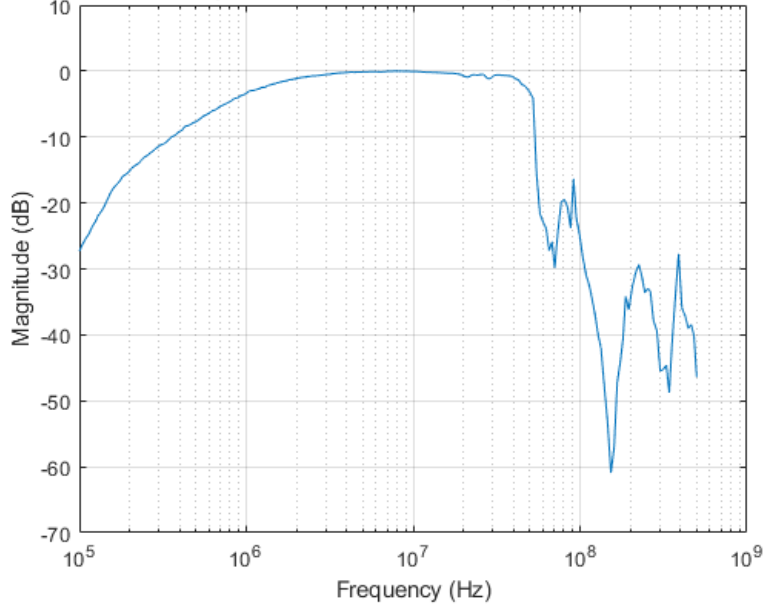


Figure R2: Normalized frequency response of the end-to-end OWC antennas used in the experimental work, in a fixed indoor configuration including reflections off walls.

From this frequency response, we can deduce that the end-to-end OWC system, including the LOS and NLOS channels in a fixed configuration, has a bandwidth at -3 dB of approximately 49 MHz, between 1.05 MHz and 50.145 MHz. This bandwidth is also relatively flat over the frequency range from 2 MHz to 35-40 MHz. Since the cutoff frequency at 1 MHz is solely a design choice (high-pass filter to cut ambient light) and is therefore flexible, we ultimately chose to use a flat communication bandwidth of 30 MHz ranging from 1 MHz to 31 MHz for our simulations.

In order to supplement this experimental data, which corresponds to only a single realization of the optical wireless channel, we also used the approach detailed in [11] for simulating the root mean square delay spread D_{RMS} along the reception plane of the $20 \times 20 \times 2.5 \text{ m}^3$ environment considered in our simulations. The resulting spatial distribution of D_{RMS} is represented in Figure R3, and shows that the root mean square delay spread remains overall below 3 ns (with maximum values logically reached close to the walls). Such a maximum value corresponds to a 50% coherence bandwidth $B_{\text{coh}} = \frac{1}{5D_{\text{RMS}}}$ of more than 70 MHz, i.e., larger than the coherence bandwidth of 31 MHz assumed in the article. Note that this value is ultimately consistent with a wall reflectivity of 0.6 (value chosen based on experimental measurements reported in [32]) and with the results reported in [11]. Consequently, we can assume that inter-symbol interference is not a strong concern for OOK in the environment considered and if the signal bandwidth is kept to 30 MHz.

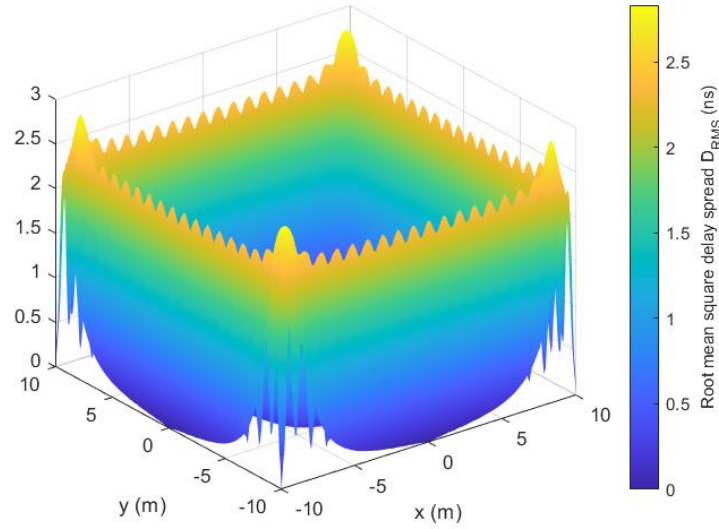


Figure R3: Spatial distribution of the root mean square delay spread D_{RMS} in the simulation environment considered in the article.

These justifications have been summarized and added to Section II-B in the revised article.

4) Comparisons across modulations are not fair:

- a. Different throughputs. OOK is capped at 10 Mb/s while m-CAP and OFDM run at 56–150 Mb/s (Table I–II, p. 8). Declaring that “OOK reaches 400 m² coverage” while others do not (Sec. III-B/C, Figs. 3–5) is not an apples-to-apples comparison. Coverage should be compared at a common data rate (or spectral efficiency) and common average optical power with the same coding assumptions. Otherwise, the paper merely confirms the obvious: lower spectral efficiency is more robust at a fixed SNR threshold.

While the data rates of the different modulations considered are indeed different, this does not mean that comparing the performance of these different modulations is not fair. Comparing coverage “at a common data rate (or spectral efficiency) and common average optical power with the same coding assumptions” would indeed have been a perfectly valid approach, but in our opinion not the only valid approach.

In this article, we have instead adopted a hardware-driven approach, considering that in practical systems, communications performance is primarily constrained by the available hardware. In our simulations, this translates into the usable bandwidth parameter, set to 30 MHz, as justified in response to the Comment 3.c from Reviewer 2 above. All other modulation design choices are then derived from this parameter, as detailed below.

Regarding OOK

We couple the OOK modulation with Manchester coding, which is quite common in OWC, in order to move most of the data signal spectrum away from DC levels and thus be able to use high-pass filtering (e.g., at 1 MHz) on reception to cut out ambient light without degrading the data signal. This results in a halving of the data rate relative to the bandwidth used, i.e., a rate of 15 Mbps (and

not 10 Mbps as stated in the first version of the article, which was a typographical error for which we apologize).

We have added these details in the waveform generation process in Section II-C.1 and the corresponding data rate value in Section III-A and in Table I (which gathers former Tables I, II and III).

Regarding *m*-CAP and OFDM

We set the remaining modulation parameters to allow for a fair comparison, i.e.:

- We set the number of useful sub-bands or subcarriers to 15 in both cases. This results in 15 transmitted sub-bands with *m*-CAP and 32 transmitted subcarriers with OFDM, due to the Hermitian symmetry and null subcarriers specific to DCO-OFDM.
- We then distributed these sub-bands / subcarriers across the entire available spectrum so as to respect the orthogonality condition for DCO-OFDM (the subcarrier spacing is thus $f_{scs} = 937.5$ kHz) and so as to ensure a sub-band width of 1 MHz and a non-null guard band between sub-bands to avoid inter-band interference.
- Each subcarrier / sub-band was then modulated with identical QAM modulations.
- A cyclic prefix (CP) of length equal to $\frac{1}{4}$ of the OFDM symbol (i.e., 8 samples) was added to each DCO-OFDM symbol to mitigate inter-symbol interference (ISI). This function is dealt with by the pulse shaping filter in *m*-CAP so that no CP is necessary with this scheme.

This results overall in a lower data rate with DCO-OFDM (from 45 Mbps to 112.5 Mbps) than with *m*-CAP (from 60 Mbps to 150 Mbps), a difference mainly explained by the CP overhead.

In order to clarify this logic of simulation and comparison, we have detailed in the new Section II-C.1 the *m*-CAP and DCO-OFDM waveforms generation process. In addition, we have added in Section III-A a paragraph summarizing the above points and in Table I the values of the different modulation parameters.

- b. BER threshold and FEC. The BER target 3.8×10^{-3} (Sec. III-A, p. 7) is imported from submarine DWDM (Ref. [30]) rather than an IM/DD OWC design with specific FEC/overheads. Using a single legacy threshold for all three modulations (including very high-order QAM) is not justified and skews the “coverage” metric.

The reviewer is correct in pointing out that the BER target of $< 3.8 \times 10^{-3}$ is not specifically defined for OWC and could thus be improved depending on the use case. Nevertheless, this threshold is commonly used in OWC literature, even if its origin could be debated. In addition, our simulations already cover many parameters, and the BER threshold defining the coverage area, even though it could be chosen differently, remains the same for all modulations and therefore allows for a fair comparison between them. For these reasons, we have chosen to keep this BER threshold in the revised version of the article.

- 5) “Misalignment” scenario and methodology: In Sec. IV-A (p. 12), “misalignment” is modeled by pointing to four points on a circle of 20 cm radius around the UE (at z separation 1.65 m), i.e., $\approx 6.9^\circ$ angular error. The coverage metric then takes the intersection across the four cases—effectively a worst-case over discrete directions. This

design is arbitrary and pessimistic. A more realistic analysis would: (i) model random angular errors with a variance that reflects tracking/IMU precision or beam quantization; (ii) report percentiles (e.g., 5/50/95%) rather than the hard intersection; and (iii) sweep misalignment statistics rather than a single 20 cm radius.

It is entirely true that the choice of the misalignment angle in the “misalignment” scenario is arbitrary and could have been done differently. For example, one of the following two approaches could have been used:

- A first approach, probably the best one, consists in developing a statistical model of the impact of misalignment on the OWC channel gain when a Lambertian source is used, similar to what is done in [R1] for a Gaussian laser source. We have actually developed such a model, but presenting and using it in detail (i.e., including all the modeling, mathematical derivations and parametric study) in the article submitted here would have resulted in a manuscript that is too long and dense, even though the study of misalignment is not the core focus of this article. For these reasons, we did not use it here.
- Another approach would have been to replicate the simulations based on the pointing radius (for example, by varying this pointing radius between a minimum and maximum value) rather than using a single fixed value. However, with the material resources at our disposal, each iteration takes approximately two days because calculating indirect paths is extremely time-consuming. A more comprehensive analysis of misalignment across the pointing radius would nevertheless involve simulating other pointing radius values, as well as other combinations of points on the same pointing circle (or even combinations of points from different circles), which would ultimately lead to very lengthy simulations. Again, given that the study of misalignment is not the focus of our article, we did not follow this method.

For these various reasons, we preferred to restrict simulations to four points on the same pointing circle with a pointing radius that we felt was both significant and not excessive, given the performance currently reported in the literature in terms of positioning and optical pointing.

The reviewer is also right to point out that defining communication coverage in “Misalignment” scenario as the intersection of the coverages according to the four pointing directions just mentioned is pessimistic. This assumption, although pessimistic, can nevertheless be justified by the fact that it is theoretically possible for the misalignment value to change during the reception of the same symbol. However, this change in misalignment can cause a drop in SNR and therefore BER such that the symbol will no longer be received correctly, thus implying a loss of communication even though the UE has not changed position. This assumption could certainly be nuanced by introducing a statistical dimension, which we have not done for the reasons mentioned above, but it is not without foundation. Furthermore, we believe it is intellectually honest to compare the ideal beamsteering case with a pessimistic misalignment case in order to bound the impact of misalignment on communication performance relative to the ideal case.

In order to better justify our definition of communication coverage in the “Misalignment” case, we have added and summarized the arguments above in Section III-C.

[R1] A. A. Farid and S. Hranilovic, "Outage Capacity Optimization for Free-Space Optical Links With Pointing Errors," *Journal of Lightwave Technology*, vol. 25, no. 7, pp. 1702-1710, July 2007.

6) Reproducibility gaps:

- a. The Schulze method requires the number of wall patches N_w , their size, and reflectivity matrices. None of this discretization detail is given, so the NLOS part is not reproducible.

All these parameters and their values have been added and commented on in Table I and Section III-A.

- b. Likewise, LED/PD bandwidths and nonlinearity are not modeled; OOK's 10 Mb/s cap is not explained; and m-CAP filter lengths (" $L_{SPAN} = 10$ ") are implausibly short for $\alpha = 0.4$ and 1 MHz sub-bands. Without these, the results cannot be replicated. (Tables I–III, p. 8; Sec. II-C, III-A.)

Regarding the bandwidth of LEDs/PDs and their non-linearity

As indicated in the response to Comment 3.c of Reviewer 2, we assume that regardless of the modulation used, the simulated OWC system transmits and receives a 30 MHz bandwidth data signal. Based on the end-to-end frequency response of the optical antennas used experimentally, shown in Figure R2, we can reasonably assume for our simulations on this frequency range that neither the LED nor the PDs introduce any nonlinearities in the frequency domain.

Furthermore, as detailed in the response to Comment 3.b from Reviewer 2, and especially with m-CAP and OFDM, the transmitted data signal is first scaled so as to drive the LED within its linear current range, meaning that this step does not introduce any non-linearities either. We recognize that these assumptions may not fully capture reality, but they nevertheless constitute a good first-order approximation that has been validated experimentally.

We have clarified these assumptions in Section II-B and in the new Section II-C.1 of the revised manuscript.

Regarding m-CAP filter parameters:

We totally agree with the observation that a filter length of 10 taps is far too short for a roll-off of 0.4 and a sub-band width of 1 MHz. However, the parameter named L_{SPAN} in the article does not correspond to the filter length in taps but to its span expressed in symbol periods. The final length of the filters, named N_{taps} in this revised article, is then deduced from L_{SPAN} and from the oversampling factor N_{ss} using:

$$N_{taps} = L_{SPAN} N_{ss} + 1.$$

In practice, we set $N_{ss} = 10$ so that $N_{taps} = 101$, which seems realistic and even fairly conservative to us. However, it is true that in the initial version of the article, we referred to the parameter L_{SPAN} as "filter length," which was very ambiguous. In this new version, we have therefore revised the notation and their descriptions (notably adding the parameter N_{taps} in Table I) in order to remove any ambiguity and clarify this point.

7) Concrete errors, inconsistencies, and places where the paper misstates results:

- a. Eq. (6) sign/structure error (critical). As noted, it produces negative powers for practical angles and contradicts the narrative (Sec. II-C, p. 7; Sec. III-C, p. 10). Must be fixed and re-simulated.

There was indeed an error in the equation, which was nevertheless only a typing error with no incidence on the simulation results. Please see our response to Comment 2 of Reviewer 2 for more details.

- b. Textual contradiction about BER vs. SNR. On p. 9 (right column, last ¶ before Fig. 3), the manuscript states: “leading to higher SNR and therefore to higher BER and coverage.” Higher SNR should lead to lower BER. This is an obvious typo but appears in a key causal explanation.

The reviewer is of course right, and the authors apologize. The error has been corrected (see Section III-B.1).

- c. PAPR argument is misapplied. The paper claims m-CAP outperforms OFDM because of lower PAPR so “for a given peak power ... average power emitted with m-CAP is higher” (p. 9). But the simulations constrain average power (via the faulty Eq. 6) and do not model peak constraints or clipping. The explanation does not match the model used.

This comment is associated with Comment 3.a from Reviewer 2. Please refer to the corresponding response for more details. In parallel, we have modified the sentence quoted by the reviewer to clarify our argument regarding PAPR (see Section III-B.1).

- d. “Monte Carlo” is a misnomer. The study appears to be a grid sweep over UE positions rather than a Monte Carlo over random channels/noises/alignments. Please either clarify or remove the term (Sec. III intro; multiple places).

The authors apologize if this was not clear. The simulations that were performed consisted of scanning the UE position along the u_x and u_y axes, which is not in itself a Monte Carlo approach. However, for each UE position, each steering configuration (‘with beamsteering’, ‘without beamsteering’, etc.) and each modulation, the simulation consists of transmitting 10^5 random bits with the modulation considered, then simulating their reception sample by sample, with Gaussian random noise added to each sample. In our opinion, this method is typically a Monte Carlo approach, hence this statement in the original article and in this revision, where we have revised the first paragraph of Section III-A to clarify this claim and the simulation process.

- e. Unrealistic transmit powers. The paper discusses 1 W average optical power and claims safety limits of >10 W at wide angles (p. 10). For 940 nm sources in indoor use, those numbers are not credible without careful IEC 62471 derivations and assumptions (exposure time, angular subtense, pupil, etc.). This undermines the physical relevance of the quantitative coverage figures.

This comment relates to Comment 2 from Reviewer 2, in the response to which we detail the method used to deduce from the IEC 62471 standard the maximum power that an LED source at 940 nm can safely emit depending on its directivity. Please refer to this response for more details.

Furthermore, we would like to point out that the formula (8) giving this maximum power contained a typographical error in the initial version of the article, for which we apologize once again. However, the simulations were carried out using the correct values.

It is true that the maximum power values obtained may seem high, especially when compared with thresholds for class 1 lasers (according to IEC 60825-1), which are most likely better known to the OWC community than thresholds for LEDs. However, in the case of a Lambertian LED with large semi-angle at half power, the optical power is radiated in a wide solid angle, which reduces the radiance. In parallel, the corresponding optical signal is not coherent, which considerably reduces its danger compared with a laser source. Consequently, the orders of magnitude of the safe power threshold differ considerably depending on whether the source is an LED or a laser, with a factor of approximately 10 at 940 nm against the laser.

However, it is also true that the practical implementation of a 940 nm LED emitter capable of delivering an optical power of more than 10 W may be problematic. It can be assumed that several LED chips could be used in parallel for this purpose, but this does not prevent the corresponding emitter from being very energy-intensive, probably bulky, and therefore ultimately unlikely to be used in practice. It is for these reasons that in the original article we considered the case where the optical power is set at 1 W regardless of the LED's directivity (in addition to the case named ' $P_t = P_{t,\max}$ ' where it was set at its maximum permitted value depending on the directivity). However, we feel that this case, although more realistic, made the article less readable while providing very little new information compared with the case $P_t = P_{t,\max}$.

That is why we have preferred to remove it from this revised version of the article, in favor of a new section on the experimental verification of the benefits of beamsteering for improving communication coverage (Section IV). In parallel, we have added a short paragraph discussing the implementation of high-power LED-based emitters in Section II-C.2, just after (8).

- f. Ambiguities in coordinates and normals. Table III puts AP at [0,0,0] (p. 8), but the room is 2.5 m high with UE a fixed z below the AP. The sign conventions for $\mathbf{n}_{AP}$ and $\mathbf{n}_{UE}$ (e.g., $\mathbf{n}_{AP} = -\mathbf{n}_{UE} = [0,0, -1]$ on p. 5) can be made clearer (is +z up or down?). These are minor but can confuse reproduction.

The Cartesian coordinate system used to describe the geometric configuration of the simulated OWC system has its origin located at the AP level, i.e. at the center of the ceiling of the 20 x 20 x 2.5 m³ room. By convention, the z-axis points upward, i.e., it points out of the room, so that any element in the room, e.g., the UE, has a negative z-coordinate. This explains why $\mathbf{n}_{AP} = [0,0, -1]$ and $\mathbf{n}_{UE} = [0,0, 1]$. To clarify all this, we have shown the Cartesian coordinate system in Figure 1 (especially Figure 1(a)) and revised its description in Section II-A.

- g. FEC threshold provenance. Using ITU-T G.975.1 (submarine DWDM, Ref. [30]) as the sole justification for $\text{BER} = 3.8 \times 10^{-3}$ in an IM/DD OWC/VLC context is not appropriate without mapping to the coding actually assumed (hard/soft decision, overhead). (Sec. III-A, p. 7.)

This comment has been addressed in the response to Comment 4.b from Reviewer 2. Please refer to that response for further details.

- h. Security angle is gestural. The paper mentions eavesdropping/privacy multiple times (p. 5, p. 15), but no secrecy metric (e.g., eavesdropper coverage area, secrecy capacity, or SNR margin) is defined or evaluated. As is, it reads speculative.

Thank you for this comment, with which we totally agree. The safety angle was mentioned solely to complement the description of the scenario of an AP having ‘sources with various fixed orientations’, but it is not the main concern of this scenario, which instead aims to evaluate the communication coverage with such a multiple fixed sources configuration rather than with a single steerable source. The issue of security deserves to be addressed separately, in much greater detail than was proposed in the initial version of the article, but this is beyond the scope of the current article.

To avoid confusion, we have removed mentions to this link security angle, simply quickly mentioning it in Section III-D when reminding the scenario of a ‘source with various fixed orientations’.

- i. Figure-related issues. Several plots do not report confidence intervals or uncertainty; axes sometimes lack clear units on legends (e.g., coverage in m^2 is stated in captions but not always on axes). A few choices (e.g., showing OOK reaching “~400 m^2 ” but omitting it from Fig. 2 entirely) make comparisons harder. (Figs. 2–5.)

The figures have been revised accordingly. In addition, please note that OOK has not been included to Figure 2 despite the comment above, because it would only add vertical bars reaching all 400 m^2 , which would not add relevant graphical information to the figure and would on the contrary make the other bars harder to distinguish by shrinking the y-axis.

8) Minor/editorial notes (non-blocking but worth fixing):

- a. Notation consistency. The manuscript alternates between $\Phi_{1/2}$, $\phi_{1/2}$, and textual “semi-angle at half-power,” and between ψ_c and “FOV.” Use one convention throughout. (Sec. II-A/B.)

The revised article has been modified accordingly.

- b. Tables I–III. Clarify L_{SPAN} (I/Q filter length = 10 taps?), CP length for DCO-OFDM, subcarrier spacing, guard nulls, and why OOK is limited to 10 Mb/s at the given bandwidths. (p. 8.)

Tables I to III have been merged into a single Table I, which has been modified along with the text in Section III-A to clarify the following points:

- L_{SPAN} refers to the filter span and not to the filter length, as written in the initial version of the article. This has been clarified with the introduction of the parameter N_{taps} which quantifies the filters length in terms of taps. Please refer to the response to Comment 6.b from Reviewer 2 for more details.
- The CP length is $\frac{1}{4}$ of the symbol duration, i.e., 8 samples. The subcarrier spacing is 937.5 kHz, as a result of the uniform distribution of the 32 orthogonal subcarriers across the 30 MHz available bandwidth. Finally, the guard nulls are the first and middle subcarriers, as

commonly done in DCO-OFDM for OWC. Please refer to the response to Comment 4.a from Reviewer 2 for more details.

- The data rate with OOK is limited to 15 Mbps (and not 10 Mbps as incorrectly stated in the initial version due to a typographical error) because Manchester encoding is used with a chip rate of 30 MHz. Please refer to the response to Comment 4.a from Reviewer 2 for more details.

c. Parameter disclosure for NLOS. Give the wall tessellation size, N_w , and convergence criteria for $(\mathbf{I} - \mathbf{H}\mathbf{G}_\rho)^{-1}$. (Sec. II-B.)

In this revised manuscript, Table I in Section III-A now includes more information about NLOS modeling.

Regarding the convergence criteria for $(\mathbf{I} - \mathbf{H}\mathbf{G}_\rho)^{-1}$, it is important to note that in our case $\mathbf{G}_\rho = \rho\mathbf{I}$ so $\mathbf{I} - \mathbf{H}\mathbf{G}_\rho = (1 - \rho\mathbf{H})\mathbf{I}$ so the convergence lies in the term $\rho\mathbf{H}$. This convergence is ensured if the maximum norm of the eigenvalues of $\rho\mathbf{H}$ is smaller than 1, i.e., if :

$$\max_i \sum_{j=1}^{N_w} |h_{ij}| < \frac{1}{\rho}, \quad (\text{R5})$$

with h_{ij} the term of the i -th row and j -th column of $\mathbf{H}$. Given that the terms h_{ij} (i.e., the LOS channel DC gains between reflectors) are typically positive and in the range of 10^{-6} and that $N_w = 4000$ in our simulations, the condition (R5) is always met so that $\mathbf{I} - \mathbf{H}\mathbf{G}_\rho$ is guaranteed to be invertible. However, we believe this discussion is out of scope of the article and have therefore not included these elements in its revised version.

d. Language. Mostly clear, but there are several tense/grammar slips and a few long sentences that can be tightened (e.g., end of p. 5; top of p. 9).

This new version of the article has been significantly revised in terms of language. The changes have not all been highlighted in blue in the track changes version so as not to obscure the major substantive changes with these formal changes, but we hope that these changes will make the article more enjoyable to read.

9) Technical and Presentation Issues:

- a. Assumption of Flat Channel: The assumption that the channel is flat (Eq. 2) and the delay spread is "minimal" is a significant simplification. For the bandwidths considered (up to 30 MHz) and in a large room (20x20m) with reflective walls, the delay spread could indeed cause Inter-Symbol Interference (ISI), especially for high-order modulations like 1024-QAM. The study would be significantly strengthened by quantifying the RMS delay spread for the simulated environment and justifying this assumption, or by incorporating a more sophisticated channel model that accounts for ISI.

This comment has been addressed in response to Comment 3.c from Reviewer 2. Please refer to that response for further details.

- b. **Vague Security Discussion:** The security angle is introduced but not explored in any depth. It is mentioned almost as an afterthought. A quantitative analysis of the "eavesdropper coverage" versus "intended user coverage" would be needed to make a substantive contribution here.

This comment has been addressed in response to Comment 7.h from Reviewer 2. Please refer to that response for further details.

- c. **Lack of Experimental Validation:** The work is entirely simulation-based. The lack of even a simple experimental setup to validate key simulation findings (e.g., the coverage shape for a fixed-orientation source) is a limitation. The authors mention future experimental validation, but its absence in the current submission weakens the impact.

This comment has been addressed in response to Comment 1 from Reviewer 2. [This revised manuscript now includes a full section \(Section IV\) dedicated to the experimental validation of the interest in beamsteering for communication coverage extension.](#) Please refer to that response for further details.